N-WAVE

Nextflow Workflow Authoring and Visualization Environment

Documentation & User Guide

November 25, 2025

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0 Preface

0.1 On the Documentation

This manual is a work in progress, if you add any functionality please add its documentation to this file and re-export it to pdf.

Downloading, installing or verifying anything mentioned in this document is a responsibility of the reader. No one else can be held accountable in case of damage caused by following the manual.

If you feel like anything is missing or there are mistakes in the documentation, please add/fix them and note changes in commits if pushing to git.

1. Installation & Setup

Installing N-WAVE can be done through any of three ways:

1.1 Manual Installation

1.1.1 Requirements

The following components are required for installation:

- MongoDB v5 or higher (you can download it here)
- Node.js v18 or higher (to check, run "$node --version" in a terminal or get here)
- pnpm or npm (to check, run "$pnpm –version" in a terminal or get here)

1.1.2 Download, Run & Stop

To download N-WAVE:

- Download N-WAVE
- Unzip the downloaded file in a location of your choice
- (Optionally rename backend/env.example to backend/.env, see 1.3)

To run N-WAVE:

Running N-WAVE manually requires running both the front- and backend separately:

Running the frontend:

- Open a terminal and navigate into N-WAVE/frontend
- (If this is your first time running N-WAVE: run $pnpm install)
- Run $pnpm dev
- You should see something like this →

```
VITE v6.3.5  ready in 447 ms

➜  Local:   http://localhost:5173/
➜  Network: use --host to expose
➜  press h + enter to show help
```

Running the backend:

- Open a terminal and navigate into N-WAVE/backend
- (If this is your first time running N-WAVE: run $pnpm install)
- Run $pnpm dev
- You should see something like this →

```
[nodemon] 3.1.10
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: ts,json
[nodemon] starting `ts-node src/server.ts`
Server running on port 5001
MongoDB Connected...
```

To stop N-WAVE simply press Ctrl+C in both terminals.

1.2 Dockerized Installation

1.2.2 Requirements

As the name suggests, Docker needs to be installed on your system.
To run N-WAVE, the Docker Engine needs to be running, so open Docker Desktop when starting N-WAVE and leave it open.

1.2.3 Download, Run & Stop

To download N-WAVE:

- Download N-WAVE by getting one of the releases from the GitHub repository
- Unzip the downloaded file in a location of your choice
- (Optionally modify the .env-file in the root directory, see 1.3)

To run N-WAVE:

- Open a terminal and navigate into the project's root directory
- Run $docker compose up -d (This will start the respective containers)

```
E:\Studium\N-WAVE\Test\N-WAVE>docker compose up -d
[+] Running 7/7
 ✔ Network n-wave_appnet         Created
 ✔ Volume n-wave_nwave_results   Created
 ✔ Volume n-wave_mongo-data      Created
 ✔ Container n-wave-frontend-1   Started
 ✔ Container n-wave-dind-1       Started
 ✔ Container n-wave-mongo-1      Started
 ✔ Container n-wave-backend-1    Started
```

To stop N-WAVE run one of the following commands from the project's directory:

- Stop N-WAVE: $docker compose stop
- Stop N-WAVE and remove all data: $docker compose down -v

```
E:\Studium\N-WAVE\Test\N-WAVE>docker compose stop
[+] Stopping 4/4
 ✔ Container n-wave-frontend-1   Stopped
 ✔ Container n-wave-backend-1    Stopped
 ✔ Container n-wave-dind-1       Stopped
 ✔ Container n-wave-mongo-1      Stopped
```

1.3 Semi-automatic Installation

The semi-automatic installation is based on the manual installation but performs a requirements check and downloads missing dependencies itself. The only thing you need to install manually is MongoDB.

1.3.1 Requirements

The following components are required for installation:

- MongoDB v5 or higher (you can download it here)

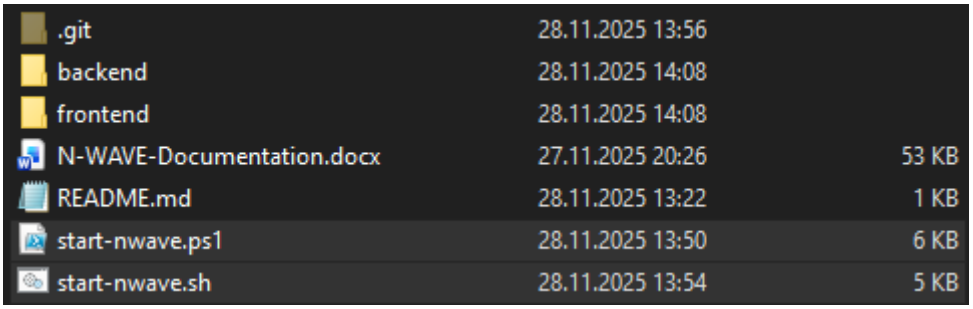
1.3.2 Download, Run & Stop

To download N-WAVE:

- Download N-WAVE
- Unzip the downloaded file in a location of your choice
- (Optionally rename backend/env.example to backend/.env, see 1.3)

To run N-WAVE, start one of the two scripts in the project's root directory:

- For Windows: right click "start-nwave.ps1" and click "Execute with Powershell"
- For UNIX: Grant execute rights once through $chmod +x start-nwave.sh , then run



.git	28.11.2025 13:56	
backend	28.11.2025 14:08	
frontend	28.11.2025 14:08	
N-WAVE-Documentation.docx	27.11.2025 20:26	53 KB
README.md	28.11.2025 13:22	1 KB
start-nwave.ps1	28.11.2025 13:50	6 KB
start-nwave.sh	28.11.2025 13:54	5 KB

To stop N-WAVE simply press Ctrl+C in the front- and backend-terminal.

2 Using N-WAVE

2.1 Accessing N-WAVE

To access N-WAVE, simply open http://localhost:5173/ through a web-browser of your choice on the device you started N-WAVE with.

2.2 Creating A Workflow

To create a new workflow, simply click “New Workflow” on the home screen of N-WAVE.

You will then see a blank canvas to which you can add nodes by pressing the “+”-Sign in the top right corner of the screen.

To return to the home screen, rename, download and run the workflow use the bottom bar.

Double-clicking a placed node opens up a menu to modify the nodes behavior.

We will now explain each node type in detail:

2.2.1 Input Nodes

Input nodes are used to provide data to the workflow. As of the moment, this can only be done through files. To do so, simply select the “File Input”-node and click on the “Upload File(s)”-button once the node appears on the canvas.

Currently all file types are accepted, however some extraordinary ones may not produce any useful results. Choosing the right filetype is left as an exercise for the user.

There is also no limit to file size or how many files you can upload to a node, but appropriate response by the application should be expected upon exceeding reasonable limits.

Note that input nodes can only be connected to other nodes once a file has been uploaded.

2.2.2 Operators

Filter:

The filter node takes in a set of files and a string of characters. It then filters the files based on the string and removes all lines where the string is not present.
Different filter modes can be selected, all of them being case-insensitive:

- Contains: Only accepts lines that contain the string at any point.
- Starts With: Only accepts lines that start with the string.
- Ends With: Only accepts lines that end with the string.
- Matches (Regex): Only accepts lines that match the regex somewhere.

Map:

The map offers two modes: Change Case & Replace Text

Change Case changes the case of all letters within the input data provided:

- Case: Uppercase – changes all letters to upper-case
- Case: Lowercase – changes all letters to lower-case

Replace Text takes in two string of characters and replaces each occurrence of the first with the second.

Merge:

Merge combines all input data provided through a separator. Only data of the same filetype shall be merged and the output file created is of the same type.

2.2.3 Core

Process:

The process node allows you to execute scripts through nextflow processes.
For more information regard the nextflow documentation on processes.

FastQC:

The FastQC node creates a quality cotrol report on sequenced reads in DNA analysis.
As input it receives a set of FASTQ files and provides three output slots:

- **FastQ**: Forwards the FASTQ input files
- **ZIP Archives**: Returns a detailed QC report as a zip archive
- **HTML Reports**: Creates a HTML document as an easy access visual report

FastQC can be configured manually through the following parameters:

- **Format**: Allows you to force input file format if it cannot be detected automatically.
- **Threads**: Enables parallel processing
- **K-mer size**: Set k for K-mer frequency analysis

The following advanced options are in place:

- **Disable grouping**: Disables grouping of bases for reads with base pair count > 50
- **Adapters file**: Set a custom adapters file path to check sequences against adapter
- **Limits file**: Set a custom limits file path to edit QC criteria

Furthermore it is possible to set a custom docker container image, its resource consumption and runtime limit.

Trimmomatic:

Trimmomatic is used to trim reads in FASTQ-files.
For that it takes in FASTQ-files and a QC report to determine trimming parameters.
It outputs 2-3 things depending on the configuration:

- **Trimmed**: A file containing all successfully trimmed reads
- **Unpaired**: Rreads from paired-end mode where the partner read was discarded
- **Logs**: Information about Trimmomatic configuration and execution

Trimmomatic can be configured manually through the following parameters:

- **Leading**: cut bases from start of read (5' end) if quality is below Leading threshold
- **Trailing**: cut bases from end of read (3' end) if quality is below Trailing threshold
- **Sliding Window**: Consists of A:B where A is the Window size and B the quality threshold, slides over read and cuts read at window start if average quality in the window drops below threshold

The following advanced options are in place:

- **Adapter file**: Contains a list of adapters (base sequences) that are present in a read unintentionally and need to be removed
- **Custom Steps**: Customize Trimmomatic further by setting more parameters
- **Phred Score**: Change how qualities are measured (for more information, see here)

Furthermore it is possible to choose from a set of docker containers for Trimmomatic execution and set resource consumption and runtime limit.

2.2.4 Output

The "Display Output"-node takes in a workflow's result and creates visual output.
It also allows for downloading all input data either through Single file download (the downloaded filetype may be changed) or through merging the output of all files to a txt.
Note that the downloaded filetype can not be changed when selecting all files for viewing.
Changing the filetype does not affect the output's content. The displayed output will also be saved as a file of the chosen filetype to the project's directory.

2.3 Importing A Workflow

As of November 2025, importing a workflow is not supported.

2.4 Executing & Monitoring A Workflow

To execute the workflow, simply click the green button in the bottom bar.
An execution panel will appear and provide information on the status of the workflow run, detailing the status of each workflow node and the overall progress.

2.5 Retrieving Workflow Results & Logs

To fetch results in case of success or logs in case of a failed run, open the project's root directory in a file explorer.

The relative paths of logs and results from the project's root are as follows:

- Results: N-WAVE/backend/results/%WORKFLOW_ID%/results
- Logs: N-WAVE/backend/results/%WORKFLOW_ID%/nextflow/.nextflow.log

where %WORKFLOW_ID% should contain the name of the workflow in question.

3 Troubles & Shoots